

Chapter 23: Generative Artificial Intelligence in Higher Education: Opportunities, Risks, and Governance

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Abstract

The rapid public availability of generative artificial intelligence (GenAI) systems, particularly large language models capable of producing fluent text, code, and images, has introduced one of the most consequential shifts in higher education since the arrival of the internet. This chapter examines the dual character of generative AI as both an instrument of pedagogical innovation and a source of institutional risk. It begins by tracing the technical evolution that made these systems usable at scale, then analyses three categories of opportunity: personalised and adaptive learning, augmentation of teaching and academic labour, and acceleration of research and administrative workflows. The chapter then turns to the principal risks, including threats to academic integrity, algorithmic bias and factual unreliability, data-privacy exposure, and the widening of equity gaps between well-resourced and under-resourced learners. Against this backdrop, the chapter argues that prohibition is neither feasible nor desirable, and that the central task for institutions is governance. It proposes a layered governance model that connects national and supranational policy, institutional strategy, and course-level assessment redesign, and concludes by identifying directions for future research. The overarching claim is that the value generative AI ultimately delivers to higher education will be determined less by the technology itself than by the quality of the human and institutional frameworks built to direct it.

Keywords: generative artificial intelligence; large language models; higher education; academic integrity; AI governance; educational technology

1. Introduction

Few technologies have moved from research novelty to mainstream adoption as quickly as contemporary generative artificial intelligence. Within months of the public release of conversational large language models, students, instructors, and administrators were experimenting with tools that could draft essays, explain dense theory, debug software, summarise scholarly literature, and converse in natural language across dozens of subjects. Higher education, an institution whose core currencies are the production and assessment of knowledge, found itself confronting a technology that appeared to produce knowledge-like outputs on demand.

The initial institutional response in many universities was defensive. Some moved to detect and prohibit the use of these tools, treating them primarily as a vector for cheating. This reaction, while understandable, mistook a symptom for the phenomenon. Generative AI is not merely a more sophisticated form of plagiarism; it is a general-purpose capability that is reshaping how text, code, and media are produced across the economy that universities prepare students to enter. A purely prohibitive stance therefore risks leaving graduates unprepared for workplaces in which fluency with these tools is becoming an expected competency.

This chapter advances three arguments. First, generative AI offers substantial and largely under-realised opportunities to improve learning, reduce unsustainable workloads, and accelerate scholarship. Second, these opportunities are inseparable from serious risks that, if unmanaged, can erode the academic standards and equity that give higher education its legitimacy. Third, the decisive variable is governance: the combination of policy, institutional strategy, and pedagogical design through which universities channel the technology toward educational ends. The remainder of the chapter develops these arguments in turn, beginning with the technical background necessary to understand both the promise and the limitations of the systems in question.

History offers a useful corrective to both panic and hype. The calculator, the personal computer, the internet search engine, and the open web each prompted predictions that they would either destroy or perfect education, and in each case the eventual reality was more nuanced: the technology became woven into practice, altered what was worth teaching and assessing, and rewarded institutions that adapted thoughtfully over those that resisted or capitulated. Generative AI is likely to follow a similar trajectory, but with a compressed timeline and a broader reach, because it operates directly on the medium of academic work itself, namely language. That difference in degree is large enough to function as a difference in kind, and it is what justifies sustained and systematic institutional attention rather than ad hoc reaction.

2. Background and Evolution of Generative AI

Generative AI refers to a class of machine-learning systems that produce new content rather than simply classifying or predicting from existing data. The current generation of these systems is built predominantly on the transformer architecture, which uses a mechanism known as attention to model relationships among elements of a sequence. Trained on very large corpora of text, code, and increasingly multimodal data, large language models learn statistical patterns of language that allow them to generate coherent continuations of a given prompt.

Two developments converted this research capability into a tool with mass adoption. The first was scale: as the number of model parameters and the volume of training data grew, models exhibited qualitatively new abilities, such as following instructions and performing tasks for which they were not explicitly trained. The second was the addition of alignment techniques, including instruction tuning and reinforcement learning from human feedback, which made model outputs more helpful, more controllable, and better matched to user intent. The combination produced systems accessible through ordinary conversation, removing the technical barrier that had previously confined such tools to specialists.

It is essential to recognise what these systems are and are not. A language model generates text by predicting plausible sequences of tokens; it does not possess understanding, beliefs, or a verified model of the world. Consequently, it can produce confident, fluent statements that are factually wrong, a failure mode commonly described as hallucination. It can reproduce and amplify biases present in its training data. And its knowledge is bounded by the data on which it was trained unless it is connected to external sources. These properties are not incidental defects to be eliminated in the next release; they are structural features that shape how the technology should be deployed in educational settings. Any responsible discussion of opportunities must therefore be read alongside these constraints.

Two further developments are reshaping these limitations in ways directly relevant to education. The first is multimodality: contemporary systems increasingly process and generate not only text but images, audio, and video, expanding their use into laboratory simulation, visual explanation, and language practice. The second is retrieval augmentation, in which a model is connected to a curated, authoritative knowledge base so that its responses are grounded in verifiable sources rather than in unaided recall. Retrieval-augmented systems can substantially reduce, though not eliminate, the problem of fabrication, and they allow institutions to anchor a tool in approved course materials. Neither development removes the need for human verification, but both change the calculus of what responsible classroom use can look like, and they signal that the technology a university evaluates today will not be the technology its students use next year.

3. Opportunities in Higher Education

3.1 Personalised and Adaptive Learning

Perhaps the most pedagogically significant opportunity is the prospect of personalised support at scale. Effective one-to-one tutoring has long been recognised as among the most powerful educational interventions, yet it has remained economically out of reach for most students. Generative AI offers a partial approximation: a tireless, patient interlocutor available at any hour, able to rephrase an explanation, generate practice problems, and adapt to a learner's pace. A student who is reluctant to ask a question in a large lecture may be willing to interrogate a model repeatedly until a concept is clear.

Used deliberately, such tools can support metacognition, prompting learners to articulate their reasoning, identify gaps, and test their understanding through dialogue. The educational benefit, however, is conditional. If a model simply supplies finished answers, it can short-circuit the productive struggle through which deep learning occurs. The design challenge is therefore to configure these tools to scaffold thinking rather than to replace it, for example by withholding direct solutions and instead posing guiding questions.

3.2 Augmenting Teaching and Academic Labour

For instructors, generative AI can absorb a meaningful share of the repetitive labour that crowds out higher-value work. It can draft lecture outlines, generate varied assessment items, produce first-pass rubrics, create worked examples at multiple difficulty levels, and translate materials into plainer language or other languages for diverse cohorts. By compressing preparation time, these tools can free academic staff to concentrate on mentorship, feedback, and the relational dimensions of teaching that machines cannot supply.

The same capability extends to feedback itself. Models can provide rapid formative comments on student drafts, allowing learners to iterate before submitting work for summative evaluation. Here too the benefit is contingent on design and oversight: automated feedback must be checked for accuracy and should complement, not substitute for, the judgement of a qualified instructor, particularly where stakes are high.

3.3 Accelerating Research and Administration

In research, generative AI assists with literature synthesis, summarisation of dense material, drafting and editing, coding and data analysis, and the generation of hypotheses to be tested. For non-native speakers of the dominant languages of scholarship, the editing capabilities can lower a long-standing barrier to

publication and participation. Administratively, the technology can streamline correspondence, support services, scheduling, and the drafting of routine documentation, reducing the bureaucratic load on staff and students alike. Across all of these uses, the recurring principle is augmentation under human judgement rather than autonomous substitution; the model accelerates a process whose direction and validation remain human responsibilities.

3.4 Accessibility and Inclusion

A frequently overlooked opportunity is the potential of generative AI to advance accessibility. For students with disabilities, these tools can generate alternative text for images, transcribe and summarise lectures, simplify complex passages, and convert material between modalities to suit different needs. For students studying in a second language, on-demand translation and rephrasing can make otherwise inaccessible content comprehensible. Used in this way, the technology can function as a form of assistive infrastructure that helps the institution meet its obligations to a diverse student body. As with every other application, the benefit depends on careful design and human oversight, since an inaccurate transcription or a distorted simplification can mislead precisely the learners it is meant to support. Nonetheless, the inclusive potential of these tools is significant and deserves a central rather than incidental place in institutional planning.

4. Risks and Challenges

4.1 Academic Integrity and Assessment

The most immediate concern is the integrity of assessment. When a model can produce a competent essay or solve a problem set in seconds, traditional unsupervised written assignments lose much of their power to certify individual learning. Detection tools have proven unreliable, generating both false positives that wrongly accuse students and false negatives that miss machine-generated work; they can also disadvantage particular groups, including non-native writers whose prose is mistakenly flagged. Reliance on detection alone is therefore an unstable foundation. The more durable response, discussed below, lies in rethinking what and how universities assess.

4.2 Bias, Accuracy, and Epistemic Risk

Because models learn from human-generated data, they inherit the biases embedded in that data and may reproduce them in ways that are difficult to detect. Their propensity to generate fluent but false statements poses a distinct epistemic hazard in an educational context, where the cultivation of accurate knowledge and disciplined scepticism is a core mission. A particular danger is automation bias, the tendency of users to over-trust authoritative-sounding outputs. Students who have not yet developed expertise are least equipped to identify a model's errors, which makes the deliberate teaching of critical evaluation an essential counterpart to any adoption of these tools.

4.3 Data Privacy and Security

Interacting with generative AI services often involves transmitting data to external providers. When students or staff enter personal information, unpublished research, or institutional data into consumer tools, that information may be stored, processed, or used in ways inconsistent with privacy obligations and data-protection law. Universities must therefore consider where data flows, what contractual and

technical protections apply, and which categories of information should never be exposed to third-party systems. Procurement decisions and clear usage guidance are central to managing this exposure.

4.4 Equity and the Digital Divide

Finally, generative AI can widen as well as narrow inequalities. The most capable systems are frequently available only through paid subscriptions, and effective use depends on reliable connectivity, suitable devices, and the literacy to prompt and evaluate outputs well. Without deliberate intervention, the technology risks conferring its greatest advantages on students who are already privileged, while leaving others further behind. Equity must therefore be treated as a design constraint and a governance priority, not as an afterthought to adoption.

4.5 Over-Reliance and the Deskilling of Learners

A subtler but profound risk is the gradual erosion of the very capacities that education exists to build. If learners habitually delegate writing, reasoning, and problem-solving to a model, they may never develop fluency in those skills themselves, much as habitual reliance on automated navigation can atrophy a person's sense of direction. The danger is acute precisely because the tool is so capable: the short-term output looks excellent, which conceals the long-term failure to learn. This concern does not argue for prohibition, since avoidance leaves students unprepared for an AI-saturated workplace, but it does argue for intentional sequencing. Learners arguably need to develop foundational competence before they are permitted to automate it, in the same way that students master arithmetic before relying on calculators. Designing curricula that distinguish between formative stages, where unaided practice is essential, and later stages, where AI augmentation is appropriate, is among the most important pedagogical tasks the technology imposes.

4.6 Environmental and Economic Costs

The training and operation of large generative models consume substantial energy and water and depend on concentrated computational resources controlled by a small number of commercial providers. For universities, this raises two linked concerns. The first is sustainability: institutions with environmental commitments must weigh the ecological footprint of large-scale AI use against its benefits. The second is dependence: building core teaching and administrative processes around proprietary external services creates strategic vulnerability to changes in pricing, availability, and terms of use. Awareness of these costs should inform procurement and encourage attention to more efficient models and, where feasible, institutionally controlled alternatives.

5. Toward a Governance Framework

If outright prohibition is unworkable and unmanaged adoption is hazardous, the path forward is governance: a coordinated set of rules, structures, and practices that direct the technology toward educational goals while containing its harms. An effective framework operates at three connected layers.

5.1 Policy and Regulatory Context

At the broadest layer, national and supranational bodies are beginning to shape the environment in which universities act. Risk-based regulatory approaches, sector guidance from international organisations, and emerging professional standards establish expectations around transparency, accountability, and the protection of fundamental rights. Institutions should align their internal policies with this evolving

landscape rather than treating governance as a purely local matter, both to ensure compliance and to draw on the accumulating consensus about responsible practice.

5.2 Institutional Strategy

At the institutional layer, universities need explicit, published positions rather than ambiguous silence that leaves students and staff to guess at the rules. A coherent strategy typically includes a clear statement of permitted, restricted, and prohibited uses; guidance on disclosure and citation of AI assistance; investment in staff and student AI literacy; attention to equitable access, for example through institutionally licensed tools; and a standing body charged with reviewing and updating policy as the technology changes. Crucially, these decisions should be made with students rather than only for them, since their lived experience of the tools is indispensable to workable rules.

5.3 Pedagogy and Assessment Redesign

The most consequential layer is the course itself. Rather than attempting to make assessment immune to AI, instructors can redesign it to remain meaningful in an environment where AI is assumed to be available. Strategies include emphasising process over product through drafts, reflections, and documented iteration; using authentic tasks tied to local context and personal experience that are difficult to outsource; incorporating oral examinations and in-class demonstrations of understanding; and explicitly teaching students to use AI critically, including the evaluation and correction of its outputs as a graded skill. Approached this way, the disruption becomes an occasion to clarify what each course is genuinely trying to develop and to assess that more directly.

5.4 Implementation and Capacity Building

A governance framework is only as effective as its implementation, and the gap between a published policy and changed classroom practice is where most institutional ambitions fail. Three conditions help to close it. The first is capacity building: sustained professional development that equips academic staff not merely to comply with rules but to redesign their teaching, since most instructors received no training in a world where these tools existed. The second is iteration: because the technology evolves rapidly, policies must be living documents subject to regular review rather than fixed pronouncements that quickly become obsolete. The third is alignment of incentives, ensuring that the considerable effort required to rethink assessment is recognised and rewarded rather than treated as invisible labour. Without attention to these enabling conditions, even a well-conceived framework will remain aspirational, and the institution will default to the unmanaged adoption it sought to avoid.

6. Future Directions

Several questions warrant sustained research. The first is empirical: rigorous, longitudinal study is needed to determine the actual effects of generative AI on learning outcomes, distinguishing genuine gains in understanding from the mere appearance of productivity. The second concerns the development of AI literacy as a discipline, including how best to teach the critical, ethical, and practical competencies that responsible use requires. The third is the design of educational tools that are pedagogically intentional rather than generic, built to scaffold thinking and to respect privacy and equity by default. A fourth is the long-term institutional question of how assessment, credentialing, and the very definition of academic skill should evolve when fluent text generation is ubiquitous. Progress on these fronts will require

collaboration among educators, technologists, policymakers, and students, and a willingness to revise practice as evidence accumulates.

7. Conclusion

Generative artificial intelligence presents higher education with a genuine dilemma rather than a simple problem to be solved. Its capabilities can deepen learning, relieve unsustainable workloads, and accelerate scholarship; the same capabilities can undermine academic integrity, propagate bias and error, compromise privacy, and entrench inequality. These outcomes are not predetermined by the technology. They will be decided by the choices institutions make. This chapter has argued that the productive response is neither uncritical embrace nor reflexive prohibition, but deliberate governance that spans policy, institutional strategy, and pedagogical design, and that places human judgement and educational purpose at the centre. Universities that treat the current moment as an invitation to clarify what they value and to redesign their practices accordingly will be best positioned to harness the technology's benefits while safeguarding the standards on which their authority rests. The defining factor, in the end, will be human.

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